

Nikhil Koshal

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EDUCATION

Indira Gandhi University

Master of Computer Applications; **CGPA: 8.9/10**

Rewari, Haryana

Aug, 2022 - June, 2024

Courses: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, Object-Oriented Programming, Artificial Intelligence, Machine Learning

SKILLS SUMMARY

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: React.js, Next.js, React Native, Redux, TanStack Query, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, FastAPI, RESTful APIs, GraphQL, WebSockets

Databases: MongoDB, PostgreSQL, Redis, pgvector

AI/ML: LangChain, LangGraph, RAG, Ollama, AI Agents, Prompt Engineering, Embeddings, TensorFlow.js, NumPy, Pandas, Sentiment Analysis

DevOps & Tools: Docker, Git, GitHub, AWS, CI/CD, Postman, Linux, Claude Code, Cursor

Concepts: System Design, OOP, DSA, Microservices, RAG Architecture, Agentic Workflows

PROFESSIONAL EXPERIENCE

MindCres Technologies

Software Engineer - AI / Full Stack

Hybrid, India

Jul, 2025 - Present

- **Point of Contact** for **Andhra Pradesh State Public Grievance System (Core Product)**
- **Led end-to-end delivery** of a state-level grievance platform handling **thousands of grievances daily**, building analytics dashboards, AI chatbots, and a performance-optimized backend.
- Engineered **LLM-powered backends** using **LangGraph, LangChain, FastAPI, RAG, and Ollama** delivering chatbots, sentiment analysis pipelines and model training workflows serving **thousands of production users**.
- Optimized **OLAP-heavy dashboard queries** through **database indexing, fact tables, materialized views, and API response caching**, reducing query latency from **~90s to 5-8s (~95% reduction)** while maintaining concurrent OLTP reliability.
- Architected an **LLM response caching layer** using **diskcache with SQLite sharding** with **background refresh every 2 hours** on new data sync, achieving **90%+ cache hit rate** — significantly reducing **LLM token costs** and serving cached responses near-instantly.
- **Won 1st place** among **116 companies** in inter-company hackathon, securing the **state Government contract** for the public grievance platform.

Codespark Labs

Junior Software Development Engineer

Delhi, India

Jan, 2025 - May, 2025

- **Resolved complex circular dependency chains** between tightly coupled business logic and UI layers, **improving modularity** and reducing build times by **30-40%**.
- **Improved frontend performance** through **memoization, code-splitting and lazy loading**, resulting in **~60% faster page load times** and a noticeably smoother user experience.

Brihat Infotech

Full Stack Developer Intern

Remote

Aug, 2024 - Nov, 2024

- **Designed real-time session service** with **secure payment flows** and **session state synchronization** over **WebSockets** using the **MERN stack**.
- Implemented **event-driven architecture** for live session management with low-latency state sync across concurrent users.

ACHIEVEMENTS

- **Genesis Award, MindCres Technologies** (*Feb, 2026*) — Recognized for **outstanding performance** and end-to-end project delivery on the state grievance platform.
- **1st Place, Inter-Company Hackathon** (*Dec, 2025*) — Won against **116 companies**, securing a state Government contract.

PROJECTS

Smart Home Automation Assistant

- Built an **IoT home automation system** using **ESP32, Node.js, React Native, MQTT, and WebSockets**, with an **NLP-based AI assistant** that parses natural-language commands into **real-time device control actions**.
- **Technologies Used:** ESP32, Node.js, React Native, MQTT, WebSockets, NLP

E-Commerce Platform with AI-based Recommendation Engine

- Built a **full-stack e-commerce ecosystem** with **Google OAuth, Maps API**, and a **TensorFlow.js-powered recommendation engine** personalizing suggestions based on **50+ behavioral signals** tracked client-side.
- **Technologies Used:** React Native, Next.js, Node.js, TensorFlow.js, MongoDB, Express